Smooth Manifolds - Self Study

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Winter Break, 2024/2025

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CAS MA 741

Abstract

This text contains my notes on Smooth Manifolds, by ___ Lee. I read this book independently, starting mid spring semester of my sophomore year, 2025. My notes largely paraphrase the text, and I work out exercises when I see fit. This book was my first exposure to Differential Geometry topics.

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Chapter 1

Smooth Manifolds

1.1 Topological Manifolds

Definition 1.1: Topological Manifold

Let M be a topological space. Then M is a *topological manifold* of dimension n or a topological n-manifold if

- M is Hausdorff;
- M is second-countable;
- For all $p \in M$, there exists a neighborhood $p \in U \subseteq M$ such that $U \cong \hat{U} \subseteq \mathbb{R}^n$ for some fixed n.

Topological manifolds are a generalization of the notion of a surface in $\mathbb{R}^n$. We write the dimension of M as $\dim M$, or sometimes M^n. The superscript is not part of the manifold's name, and simply denotes it's dimension.